

FoodHub

Exploratory Data Analysis

Python Foundations

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Executive Summary



Conclusion

- **High demand on weekends:** 71% of orders occur on weekends, with **American cuisine** being the most popular.
- **Consistent food preparation efficiency:** Most restaurants show stable food preparation times (20–35 mins), indicating that **delays are not restaurant-driven**.
- **Delivery is the bottleneck:** Delivery times are longer and more variable—especially on **weekdays**, where the mean time increases to **28 minutes**.
- **Customer ratings skew positive**, but **736 unrated orders** highlight a gap in feedback.
- **No significant correlation** between cost of the order, food preparation time, and delivery time. These areas can be optimized independently.
- **4 restaurants** meet criteria for a promotional campaign (>50 ratings & avg. rating > 4.0).
- Net revenue across all orders is **~\$6,166.30**, with **29.24% of orders** cost above \$20.

Executive Summary



Actionable Insights

- **Faster delivery** is associated with **higher customer ratings**, unlike cost of the order or food preparation time.
- **Thai** cuisine which has one of the highest median food preparation time and widest IQR (i.e., highly variable) may require **optimizations in kitchen operations**.
- **Korean and Vietnamese** cuisines with **efficient food preparation and low-costs** are ideal for express-delivery promotions.
- **Weekday delivery inefficiencies** are likely due to operational constraints like traffic jams or less drivers.
- Top 5 customers and restaurants can be **targeted for loyalty programs or retention campaigns**.

Executive Summary



Recommendations

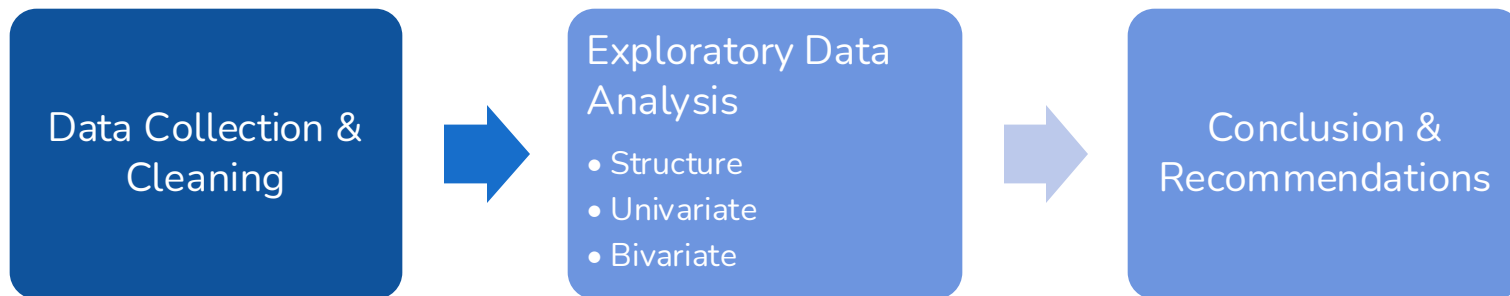
- **Optimize delivery time**
 - Hire more drivers, especially on weekdays and peak hours.
 - Use traffic-aware routing to minimize traffic and delivery delay.
 - Investigate and resolve reasons for the 10.54% of orders taking more than 60 minutes.
- **Encourage customer feedback**
 - Use small incentives like coupons and/or app-nudging notifications to promote rating participation and gather more accurate insights.
- **Employ targeted marketing by cuisine**
 - Highlight fast and affordable cuisines such as Korean and Vietnamese for express delivery promotions.
 - Promote premium cuisines (like French and high-cost Mediterranean) for quality-focused customer segments.
- **Increase customer satisfaction and retention**
 - Top 5 customers can be targeted for loyalty programs (provide benefits such as bundle discounts, free delivery, and more).
- **Promote top 4 performing restaurants** to build brand trust and increase orders.
- **Collect more granular data on the days of the week** for more specific actionable insights to enhance customer experience and boost revenue.

Business Problem Overview and Solution Approach

Problem Statement

FoodHub, an online food aggregator in New York, seeks to enhance its business performance by analyzing data from its customers' orders. The objective is to decipher and understand customer preferences, restaurant (cuisine) demand trends, and improve service efficiency to optimize operations and enhance user experience.

Methodology



Data Overview: Structure of the Data

- The data includes different attributes related to each order (illustrated in Table 1).
- There are 1898 rows (order entries) and 9 columns (variables) in the data frame.
- In terms of data types, the dataset includes 4 object columns, 4 integer columns, and 1 float column.

Table 1. Data type of each order attribute.

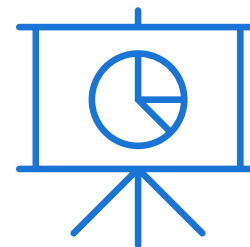
Column Name	Data Type
order_id	int64
customer_id	int64
restaurant_name	object
cuisine_type	object
cost_of_the_order	float64
day_of_the_week	object
rating	object
food_preparation_time	int64
delivery_time	int64

Data Overview: Structure of the Data

- There are **no missing values** in the data frame. However, some orders were unrated, marked as 'not given' in the rating column.
- Table 2. **Minimum, average, and maximum food preparation time.**

	Time (minutes)
Minimum	20
Average	27
Maximum	35

- **736 orders** were **not rated**, indicating potential **gaps in feedback collection**.

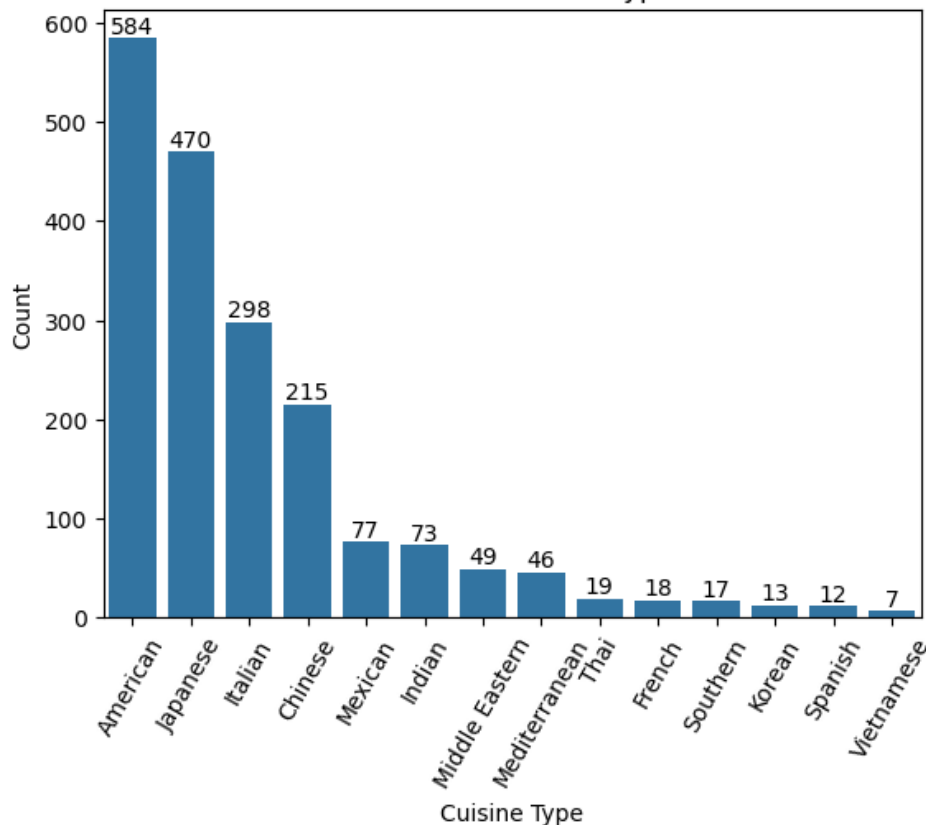


Univariate Analysis Overview

- **order_id:** FoodHub has 1898 orders in its dataset.
- **customer_id:** There are 1200 customers recorded.
- **restaurant_name:** There are 178 restaurants listed in FoodHub.
- **cuisine_type:** FoodHub offers 14 different types of cuisine to its customers.
- **cost_of_the_order:** There is a wide spread of order costs ranging from \$5 to \$35, indicating a diverse customer base and menu offerings.
- **day_of_the_week:** Peak customer activity occurs on weekends, likely due to increased leisure time at home and dining preferences.
- **rating:** The distribution is positively skewed toward higher ratings, which reflects good overall customer satisfaction—but the large "Not given" rating identifies a gap in feedback collection.
- **food_preparation_time:** Food preparation times are evenly distributed between 20 and 35 minutes. Data suggest that kitchen processes of the restaurants are consistent and optimized, thus, are unlikely cause of delivery delays.
- **delivery_time:** Delivery time show greater inconsistency and variability compared to food preparation time. Reducing delivery time and variability and could significantly improve overall order experience and satisfaction.

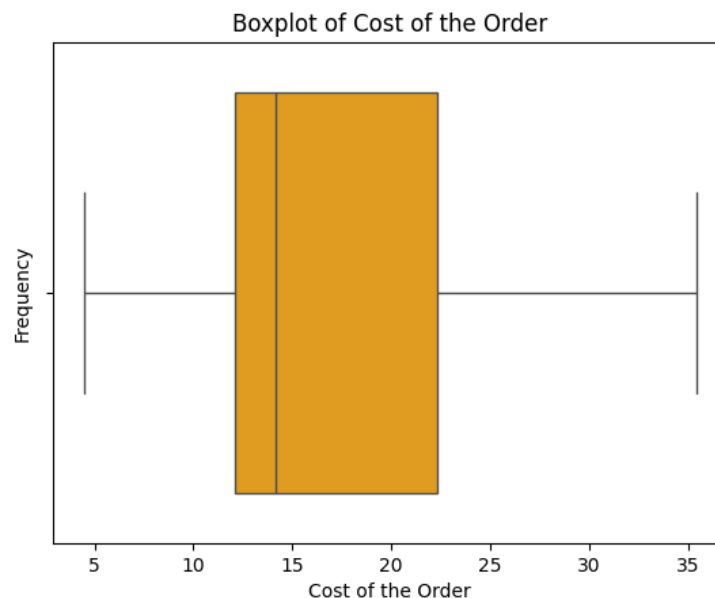
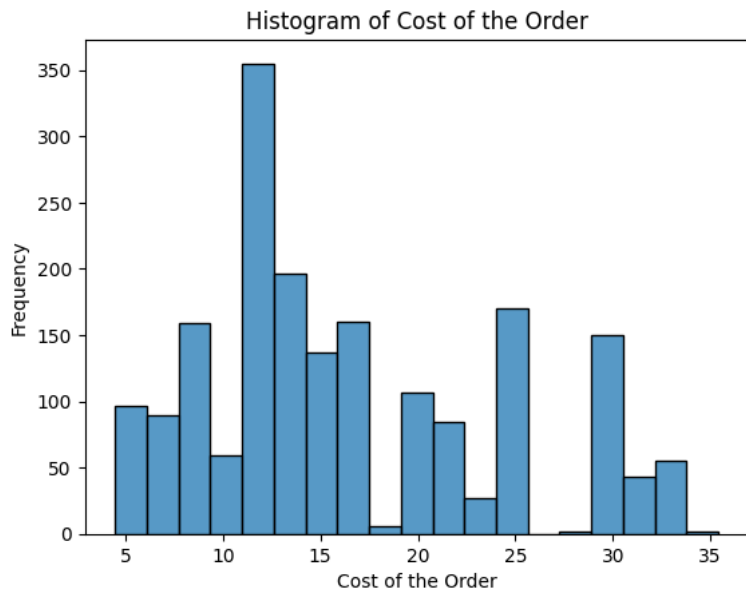
Univariate Analysis: Cuisine Type

Bar Plot of Cuisine Type



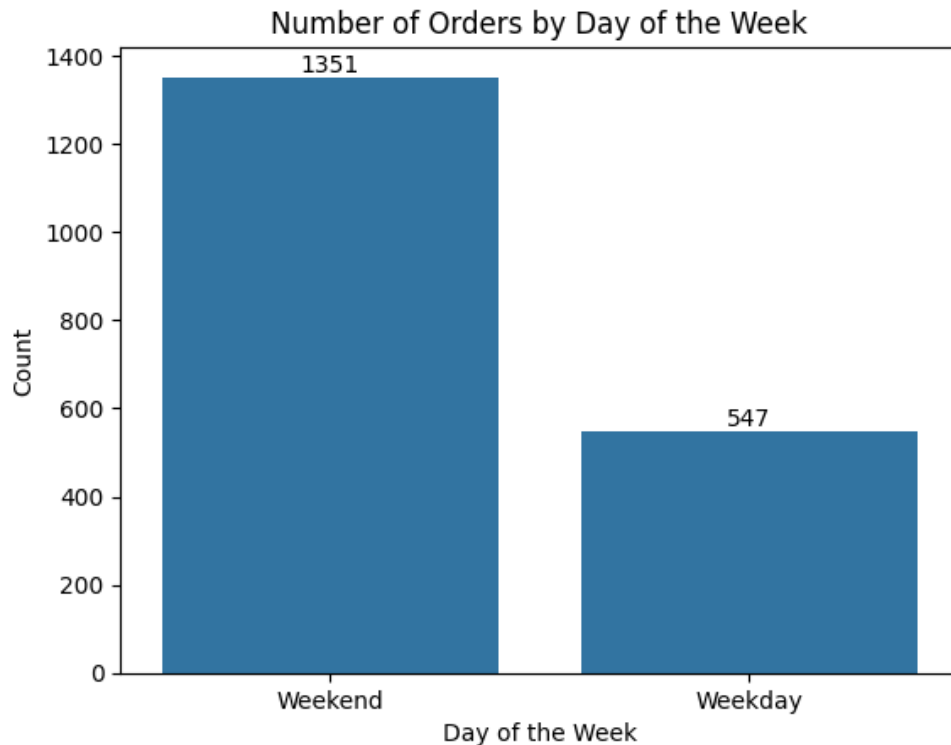
- **American cuisine dominates** the FoodHub platform, accounting for the highest number of orders (584).
- **Japanese (470) and Italian (298)** cuisines are the second and third most popular choices.
- **Chinese (215)** cuisine also has a strong presence, with significantly higher orders than Mexican.
- **Mexican (77) and Indian (73)** cuisines have moderate orders.
- **Middle Eastern (49) and Mediterranean (46)** cuisines demonstrate lower demand but still noticeable.
- **Thai, French, Southern, Korean, Spanish and Vietnamese** show minimal orders (<20), indicating underrepresentation or niche appeal.

Univariate Analysis: Cost of the Order



- The **most frequent order cost** falls in the **\$11–12** range, as seen in the histogram's peak.
- The cost distribution is **multi-modal**, indicating varying price segments—likely due to different cuisine types or order sizes.
- The **majority of orders cost between \$10 and \$25**, with a **median around \$14**.
- While no significant outliers are present, there is a **wide spread** of order costs, ranging from **\$5 to \$35**.
- The variety in pricing suggests a **diverse customer base and menu offerings**, opening possibilities for targeted pricing strategies or bundling offers.

Univariate Analysis: Day of the Week



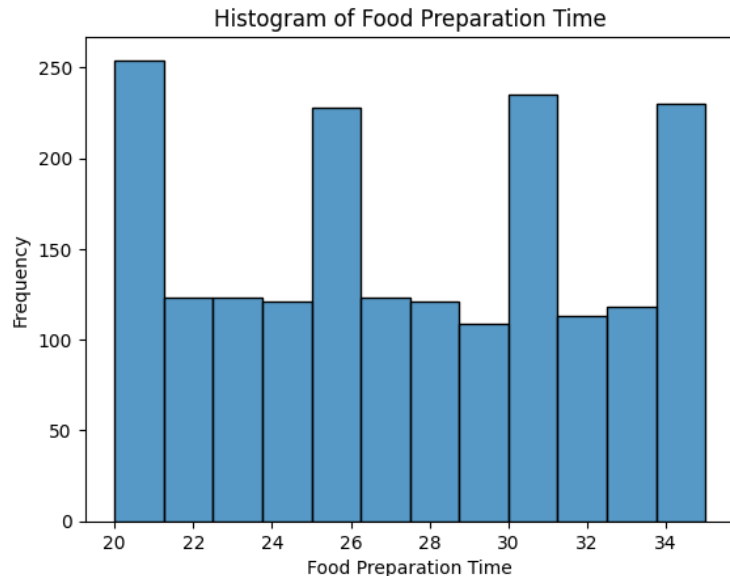
- There are significantly **higher volume of orders on weekends (1351)** than on weekdays (547) in FoodHub.
- It suggests **peak customer activity occurs on weekends**, likely due to increased leisure time at home and dining preferences.

Univariate Analysis: Rating

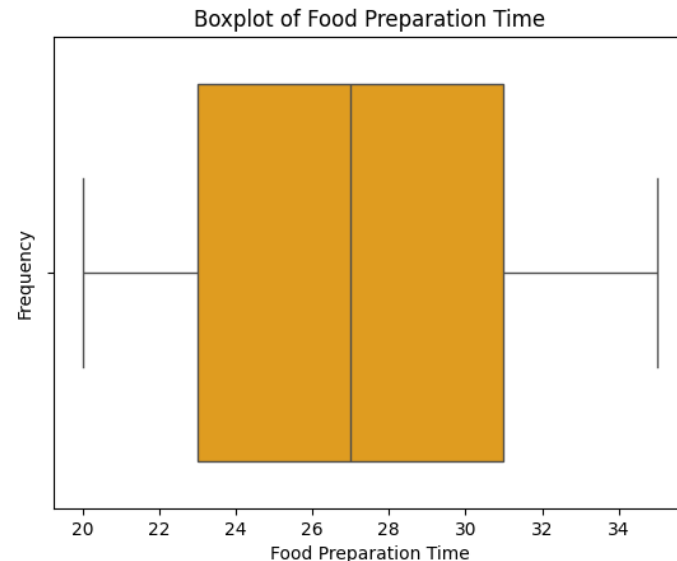


- A significant number of orders were not rated (736), demonstrating a **gap in feedback collection**.
- Most of the customers who rated the service gave a **5**, indicating a **sizeable segment of highly satisfied customers (588)**.
- **386 orders were rated 4**, while **188 orders were rated 3 (lowest count)**, indicating a smaller share of moderately satisfied customers.
- Nobody rated the service 1 or 2, suggesting fairly satisfied customers overall.
- The distribution is **positively skewed toward higher ratings**, which reflects **good overall customer satisfaction**—but the large "Not given" category should be addressed.

Univariate Analysis: Food Preparation Time

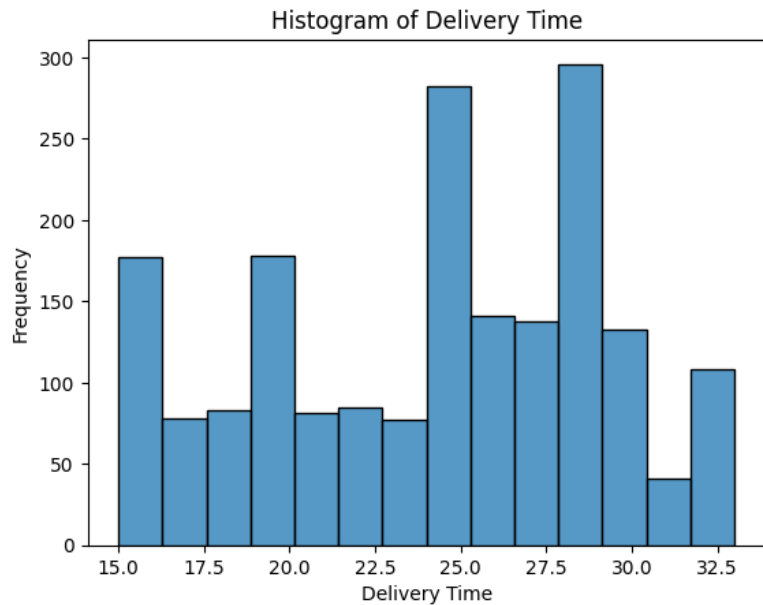


- Food preparation times are **evenly distributed** between **20 and 35 minutes**.
- Peaks are observed around 20, 24, 30, and 35 minutes indicate **batch operations** in **5-minute intervals**.

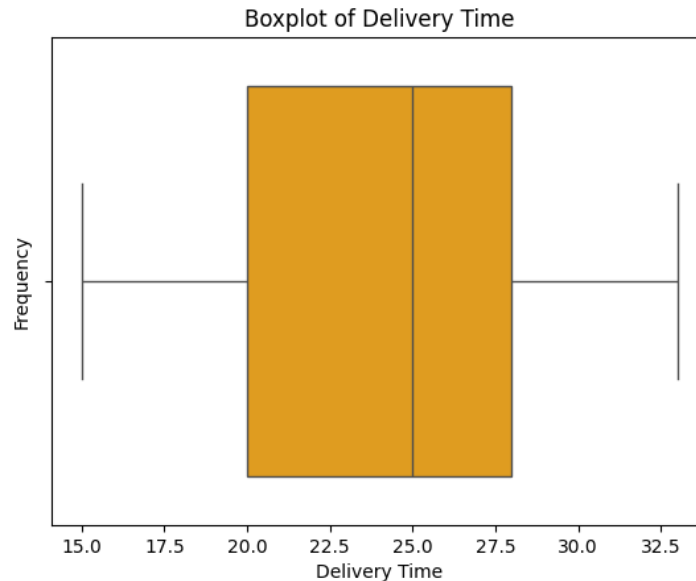


- **Majority** of orders take between **23 and 31 minutes (min)** to be prepared.
- **Median** food preparation time is around **27 min**.
- **No outliers** suggest **consistent optimized kitchen processes** of the restaurants.

Univariate Analysis: Delivery Time



- Unlike the even distribution of food preparation time, the histogram reveals **multiple peaks of variable height**.
- This suggests **inconsistent delivery times** possibly due to traffic depending on the time of the day and delivery staff availability.



- **Median delivery time** is around **25 min** with most of the orders delivered between **20 and 28 min**.
- **No extreme outliers**, but the variability suggests opportunities for **logistical optimization**.
- Overall, delivery time show **greater inconsistency** compared to food preparation time.

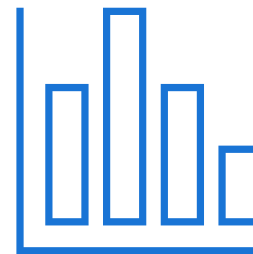
Univariate Analysis

- Table 3. **Top 5 restaurants** in terms of the **number of orders** received.

Restaurant Name	Number of Orders
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

- The most popular cuisine on weekends is **American**.
- The percentage of orders above \$20 is **29.24 %**.
- The mean delivery time for this dataset is **24.16 minutes**.
- Table 4. **Top 5 most frequent customers**.

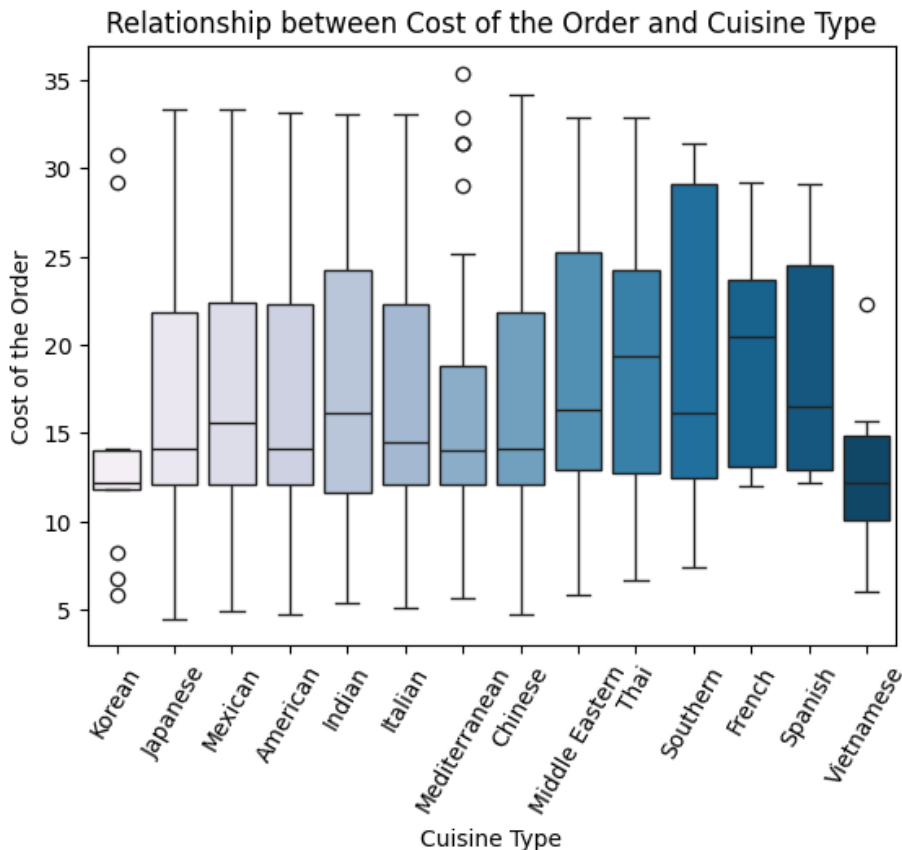
Customer ID	Number of Orders
52832	13
47440	10
83287	9
250494	8
259341	7



Multivariate Analysis Overview

- **Cuisine vs Cost of the order:** Korean and Vietnamese cuisines stand out as most affordable, whereas French and Thai cuisines have more premium costs.
- **Cuisine vs Food Preparation Time:** Korean, Vietnamese, French, Southern and Mexican cuisines demonstrate efficient kitchen turnaround while Thai, Italian, and Japanese cuisines take the longest to prepare.
- **Day of the Week vs Delivery Time:** Despite lower volume of orders, delivery time are longer on weekdays, potentially due to traffic congestion, fewer delivery personnel, or other operational inefficiencies.
- **Rating vs Delivery Time:** Faster deliveries seem to correlate with higher ratings.
- **Rating vs Food preparation Time:** No strong correlation between rating and food preparation time.
- **Rating vs Cost of the Order:** A mild positive correlation between order value and customer satisfaction is present, though not definitive due to high and overlapping error bars.
- **Correlation among variables:** No observed linear correlation between delivery time, food preparation time and cost of the order.

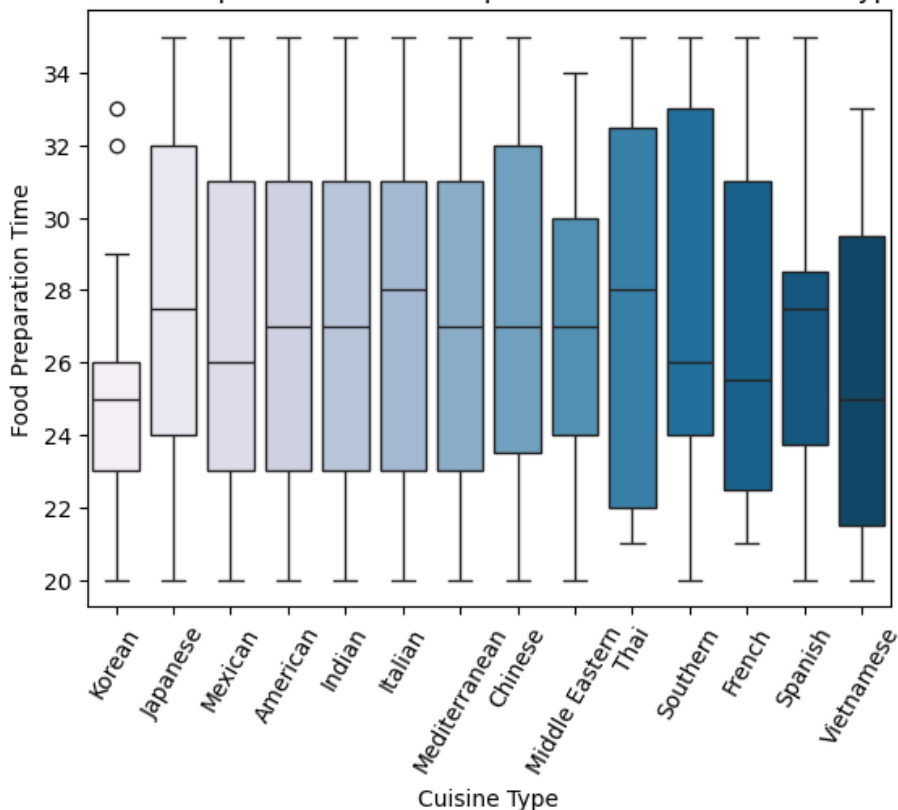
Multivariate Analysis: Cuisine vs Cost of the order



- **Korean and Vietnamese** have the **lowest median costs** and **narrower interquartile (IQR) ranges**, reflecting more consistent and affordable pricing.
- **French and Thai** cuisines are on the more expensive end with **higher median costs** and **positively-skewed toward higher costs**.
- **Southern cuisine** has the highest variation in cost with the largest IQR.
- **Japanese, Mexican, American, Italian and Chinese** cuisines show moderate median costs with broader variability.
- Despite the lower overall costs of **Mediterranean** cuisine (relatively low median cost and narrow IQR), it has several extremely high-cost outliers indicating the niche high-end Mediterranean restaurants
- **Vietnamese and Korean** also display **high-value outliers**, likely due to premium menu items or large orders.

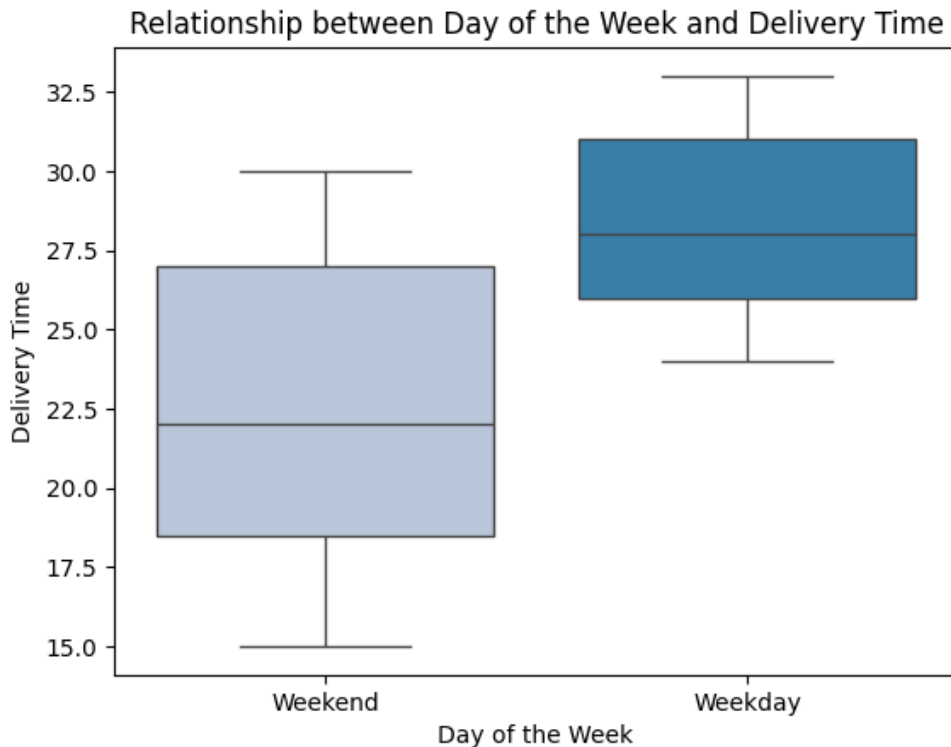
Multivariate Analysis: Cuisine vs Food Preparation Time

Relationship between Food Preparation Time and Cuisine Type



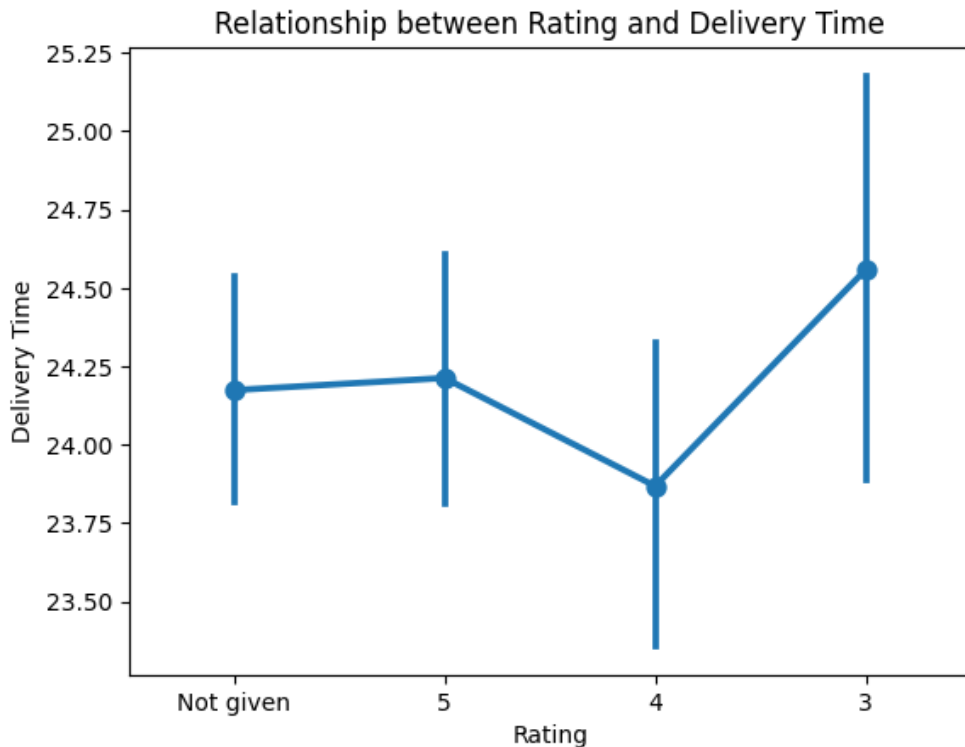
- **Korean** cuisine has the shortest median food preparation time and a tight IQR, suggesting quick and consistent preparation (prep).
- This is followed by **Vietnamese, French, Southern** and **Mexican**, indicating faster kitchen turnaround although with higher variability (wider IQRs).
- **Thai, Italian, and Japanese** cuisines take the longest to prepare as indicated by higher median values. **Thai** cuisine has the widest IQR, indicating both longer and more variable prep durations.
- Most other cuisines, such as **American, Indian, Mediterranean and Chinese**, fall into a **mid-range prep time (around 26 to 28 mins)** with moderate variability.
- **Outliers** are minimal, with only **Korean** showing a few high preparation time outliers, likely due to occasional large, specialty orders or more expensive restaurants.

Multivariate Analysis: Day of the Week vs Delivery Time



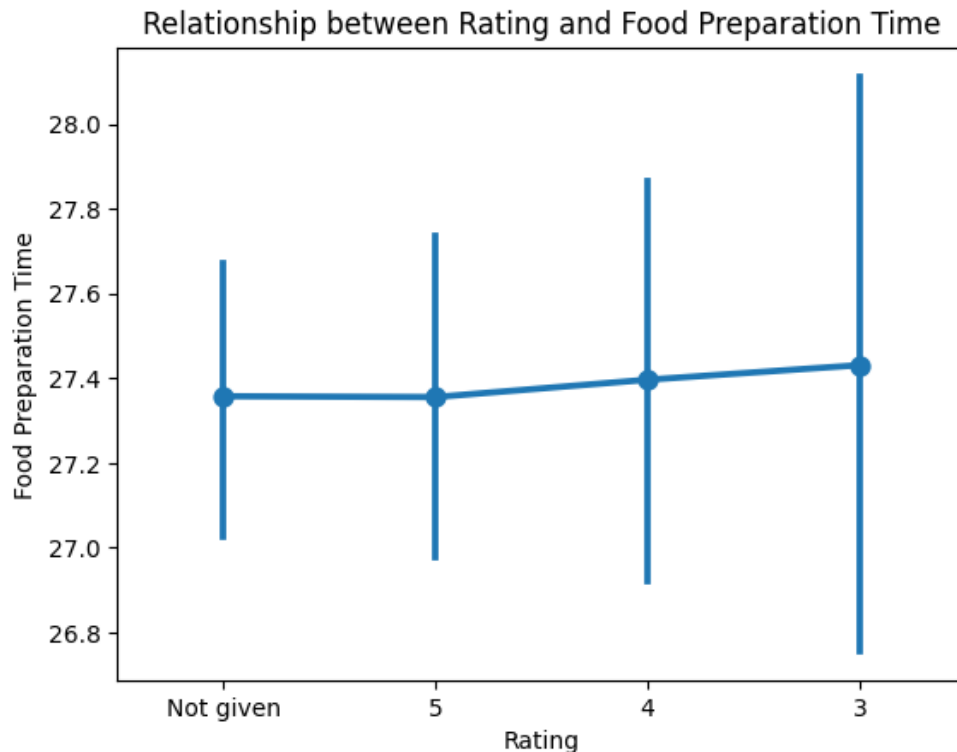
- **Median delivery time is higher on weekdays**, suggesting that orders placed during the week generally **take longer to deliver** than those on weekends — likely due to traffic congestion, fewer delivery personnel, or other operational inefficiencies.
- Despite the lower median delivery time on **weekends**, the IQR and whiskers are wider, showing **more variability in the service**.
- Larger variability on weekends can be explained by the **higher volume of orders** (data points) recorded on weekends (1351) than on weekdays (547) (found in the univariate analysis).
- **Reducing delivery variability** could significantly improve overall order experience and satisfaction.

Multivariate Analysis: Rating vs Delivery Time



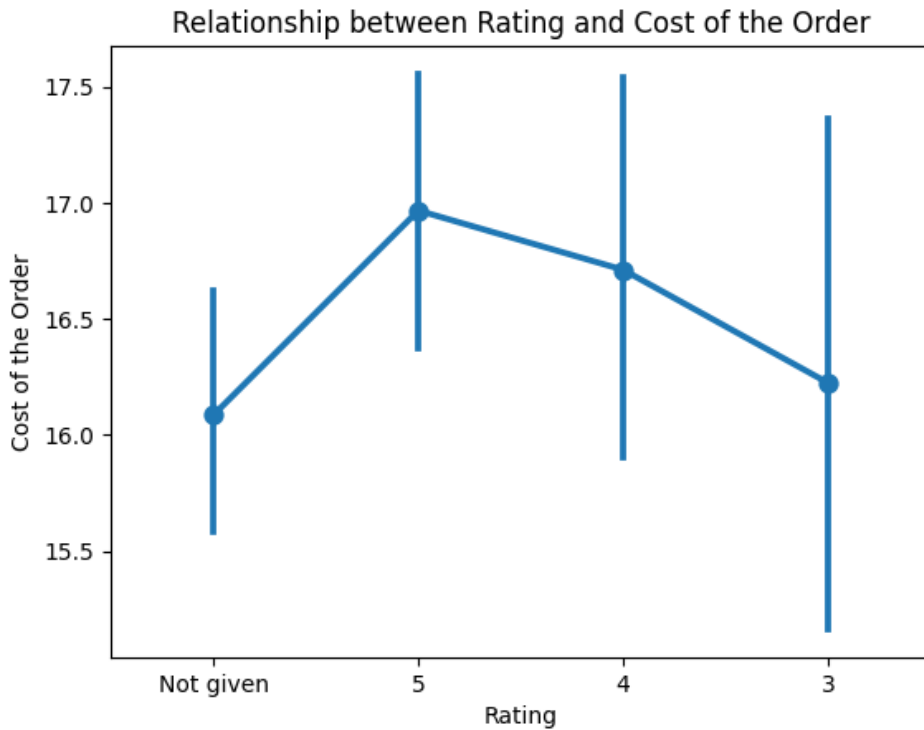
- **Faster deliveries seems to correlate with higher ratings** – Rating 4 has the **lowest average delivery time**. This also suggests that other factors like food quality can impact the rating.
- **Rating 3 shows the highest delivery time**, suggesting longer waits may impact customer satisfaction.
- **Unrated orders ('Not given')** have delivery times similar to 5-star ratings – missing ratings may not reflect dissatisfaction.
- **Delivery time generally decreases** as customer ratings increase.
- **High variability** in delivery time is observed for lower ratings, especially rating 3.

Multivariate Analysis: Rating vs Food preparation Time



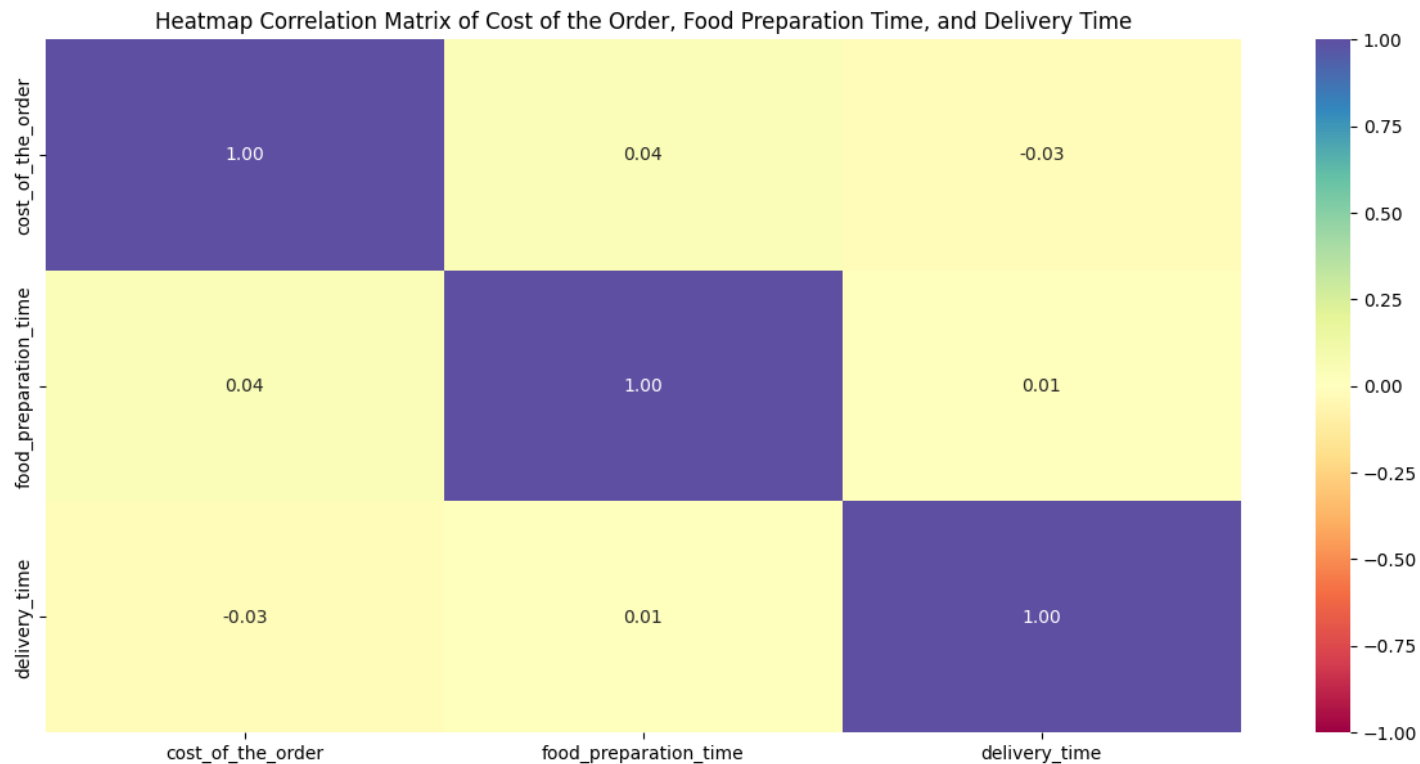
- Average food preparation time remains **fairly consistent** across all rating categories, hovering around **27.3 to 27.4 minutes**.
- A **slight increase** in preparation time is observed as the rating decreases from 5 to 3, but the change is **minimal and insignificant, especially given the considerable overlapping of error bars**
- Customers who didn't leave a rating ("Not given") also experienced similar preparation times.
- Overall, food preparation time does **not appear to significantly influence customer ratings**. This suggests that other factors (such as delivery time or food quality) may have a stronger impact on customer satisfaction.

Multivariate Analysis: Rating vs Cost of the Order



- **Higher-rated orders** (rating 5) tend to have **slightly higher average costs** (peak around \$17).
- This suggests **higher customer satisfaction** of those who spend more money — likely due to ordering higher quality meals or multiple items.
- Lower ratings (rating 3) and no ratings ('Not given') tend to have slightly lower average costs (around \$16 to \$16.3).
- **Rating 3** demonstrates **highest variability** in cost (\$15.0 to \$17.4).
- Nonetheless, **differences across rating groups may not be significant** due to overlapping error bars.

Multivariate Analysis: Correlation among variables



- **No observed linear correlation** between the variables.
- Food preparation time and delivery time **operate independently**.
- High-cost orders are **not necessarily** prepared or delivered slower (or vice-versa).

Multivariate Analysis

- **4 restaurants** meet the promotional offer criteria

- More than 50 ratings
- Average rating > 4.0

Table 5.

Restaurant Name	Number of Ratings	Average Rating
The Meatball Shop	84	4.51
Blue Ribbon Fried Chicken	64	4.33
Shake Shack	133	4.28
Blue Ribbon Sushi	73	4.22

- The **net revenue** generated by the company across all orders is around **\$6,166.30**
- **Percentage** of orders that took longer than **60 minutes**: **10.54 %**
- The **average delivery time** on **weekdays** is around **28 minutes**, while it takes around **22 minutes** on **weekends**.



Thank you for your attention!



Happy Learning !

